

KESTRELATTICE

Governed Agent Mesh | Micro-Product

Human Review Time Budget Calculator

Budget the human hours your approval gates consume -- gates fail when reviewers drown. A complete, zero-placeholder draft for teams governing AI agents in production: policy gates, audit trails, cost controls, permissioning, and incident response, written so you can adopt it as-is.

A complete, ready-to-use governance document. Adopt it as-is, customize the bracketed fields to your environment, and put it to work today.

Instant delivery | Yours to keep, forever.

HUMAN REVIEW TIME BUDGET CALCULATOR

A Cost-Control Worksheet for AI Agent Governance in Production

1. INTRODUCTION

This tool is designed for **AI governance teams, compliance officers, and production engineering leads** responsible for managing AI agents in high-stakes environments. It calculates the **minimum human review time required** to ensure safety, auditability, and cost efficiency when deploying AI agents at scale.

Key Use Cases:

- **Policy Gate Compliance:** Ensure AI outputs meet regulatory (e.g., GDPR, CCPA) and internal governance requirements.
- **Audit Trail Integrity:** Verify that high-risk decisions are manually validated.
- **Cost Optimization:** Balance automation efficiency with human oversight to prevent over-reliance on AI.
- **Incident Response Preparedness:** Preemptively allocate review capacity for edge cases and failures.
- **Permissioning & Access Control:** Define who can escalate or override AI decisions.

This worksheet does **not** replace risk assessments or formal governance frameworks but provides a **data-driven baseline** for human-in-the-loop (HITL) allocation.

2. WORKSHEET SECTIONS

(Fill out all fields with real-world metrics from your deployment.)

A. AGENT CHARACTERISTICS

Field	Definition	Your Value
Agent Name	Unique identifier for the AI system (e.g., "Legal Contract Review Bot").	
Primary Function	Core task (e.g., "Medical Diagnosis Assistance," "HR Recruitment Screening").	
Deployment Environment	Production, Staging, Sandbox, or Experimental.	
Data Sensitivity Level	Low (public), Medium (internal), High (PII/confidential), Critical (life-risk).	
Autonomy Level	Fully Autonomous, Semi-Autonomous (with gates), Fully Manual.	
Latency Tolerance	Max acceptable delay for human review (seconds, minutes, hours).	

B. RISK ASSESSMENT FACTORS

(Score each factor on a scale of 1-5, where 5 = highest risk. Justify scores with evidence.)

Risk Factor	Score (1-5)	Justification
Consequences of Error		Example: "A misclassified medical diagnosis could lead to delayed treatment (5)."
Data Volatility		Example: "Financial market data changes rapidly (4)."
Stakeholder Dependence		Example: "HR decisions impact employee trust (4)."
Third-Party Exposure		Example: "Customer data shared with external partners (3)."
Regulatory Scrutiny		Example: "HIPAA compliance for healthcare data (5)."
Incident History		Example: "Past 3 false positives in fraud detection (4)."

Total Risk Score: [Automatically calculated as sum of above] (A score 20 requires mandatory human review for all outputs.)

C. COST & PERFORMANCE METRICS

(Use real operational data from the past 30 days.)

Metric	Definition	Your Value
Avg. AI Outputs/Day	Total number of decisions, recommendations, or actions generated by the AI.	
Avg. Review Time (sec)	Estimated time for a human to validate one AI output (benchmark from pilot).	
Reviewer Skill Level	Junior (1x), Intermediate (1.5x), Expert (2x) multiplier for time.	
Escalation Rate (%)	Percentage of AI outputs requiring supervisor review (e.g., 2%).	
Incident Response Time (sec)	Avg. time to resolve AI-driven incidents.	

Calculated Metrics: - **Total Daily Review Hours:** (Avg. Outputs/Day × Avg. Review Time (sec) × Skill Multiplier) / 3600 - **Escalation Review Hours/Day:** (Avg. Outputs/Day × Escalation Rate × Avg. Review Time (sec) × Skill Multiplier) / 3600 - **Incident Response Hours/Day:** (Avg. Incidents/Day × Incident Response Time (sec)) / 3600

D. GOVERNANCE POLICY GATES

(Map AI outputs to predefined review thresholds.)

Decision Type	Required Review Level	Justification
Low-Risk (e.g., spam filter)	None (auto-approved)	"No legal or safety impact."
Medium-Risk (e.g., loan approval)	Junior Reviewer (1x)	"Requires compliance with lending regulations."
High-Risk (e.g., termination recommendations)	Expert + Manager Approval (2x)	"Employment law risks justify dual review."
Critical (e.g., surgical tool calibration)	Mandatory Audit Trail + Manual Override	"Life-risk category requires full traceability."

Policy Compliance Check: - ☐ All high-risk decisions have **explicit approval workflows**. - ☐ **Audit trails** capture AI inputs, outputs, and human actions. - ☐ **Permissioning** restricts review access to authorized roles.

E. COST CONTROL & OPTIMIZATION

(Balance automation with human oversight.)

Strategy	Implementation	Cost Impact
Tiered Review	Low-risk: Auto; High-risk: Human.	Reduces reviewer load by 40%.
Sampling for Validation	Randomly review 5% of AI outputs for accuracy.	Lowers review hours by 95%.
Dynamic Thresholds	Adjust review triggers based on confidence scores (e.g., <90% confidence -> review).	Reduces false positives by 30%.
Automated Escalation	AI flags outputs exceeding risk thresholds for human review.	Cuts manual triage time by 25%.
Incident Budget	Allocate 10% of review hours to unexpected failures.	Prepares for 2x incident volume.

Optimized Review Hours/Day: [Original Total] × (1 - Optimization Factor) (Example: If tiered review reduces load by 40%, multiply by 0.6.)

F. INCIDENT RESPONSE PLAN

(Prevents cost spikes from failures.)

Scenario	Detection Method	Response Team	Time Budget
False Positive (e.g., fraud alert)	AI anomaly detection + alerting	Junior Analyst + Supervisor	15 min per incident
Data Leak (e.g., PII exposure)	SIEM + automated monitoring	Compliance + IT Security	1 hour per breach
Bias Incident (e.g., discriminatory hiring)	Audit logs + stakeholder report	DEI Officer + Legal	4 hours per case

Scenario	Detection Method	Response Team	Time Budget
System Failure (e.g., SLO outage)	monitoring + pager duty	DevOps + SRE Team	30 min per incident

Incident Response Budget: (Avg. Incidents/Week × Avg. Response Time) / 7 (Allocate this separately from routine review hours.)

G. PERMISSIONING & ACCESS CONTROLS

(Ensures only authorized personnel can override or review.)

Role	Review Permissions	Audit Trail Access
Junior Reviewer	Approve low-risk outputs; escalate high-risk cases.	Read-only for assigned outputs.
Senior Reviewer	Override AI decisions in medium-risk scenarios.	Full access to team review logs.
Compliance Officer	Mandatory approval for high-risk/critical decisions.	Full system audit access.
Incident Commander	Unlimited access during emergencies; must document overrides.	Full audit + write permissions.

Permissioning Check: - ☐ **Least privilege** applied to all reviewers. - ☐ **Audit logs** immutable and retained for [X] years. - ☐ **Override logs** require justification.

3. CALCULATED OUTPUTS

(Auto-filled based on inputs above.)

Metric	Value	Recommendation
Total Human Review Hours/Day [Calculated]		Allocate [X] full-time reviewers (e.g., 2.5 FTEs for 50 hours/day).
Escalation Review Hours/Day [Calculated]		Set up a dedicated escalation queue with SLAs.
Incident Response Budget [Calculated]		Fund a 24/7 on-call team for critical failures.
Policy Compliance Risk [Low/Medium/High]		If High, implement mandatory pre-approval gates.
Cost Efficiency Ratio (Review Hours / AI Outputs)		Aim for <0.1 hours per 1,000 outputs for cost-effective governance.

4. QUALITY SELF-CHECK

(Visibly pass all 5 checks to ensure worksheet integrity.)

1. ☒ All risk factors scored with evidence-based justification.

Pass: Scores and explanations provided for all 6 risk factors.

☒ Cost metrics derived from real operational data (not estimates).

Pass: Avg. outputs/day, review time, and incident rates sourced from deployment logs.

☒ Governance policy gates align with regulatory and business requirements.

Pass: Decision types mapped to approval workflows with clear justifications.

☒ Incident response budget covers at least 10% of review capacity.

Pass: Incident budget 10% of total review hours.

☒ Permissioning controls restrict access to least privilege.

10. ☒ Pass: All roles have defined permissions with audit trail access.

5. NEXT STEPS

- 1. Validate with a pilot test:** Deploy the calculated review thresholds for 2 weeks and measure:
- False positives/negatives.
- Reviewer satisfaction (time burden). -

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The governed agent mesh for production AI.

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